

DHIYANESH SIDHAIYAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 7+ years building scalable cloud-native applications and microservices on AWS and GCP. Strong across full-stack development, AI/ML and LLM-powered systems, and agile delivery. Shipped 7+ mission-critical systems spanning credit, risk, and payment operations at 87% AI prediction accuracy and 100% payment automation.

TECHNICAL SKILLS

Backend: Python, Node.js, JavaScript, TypeScript, Flask, FastAPI, Express.js, Socket.IO

Frontend: React.js, Angular, Vue.js, HTML5, CSS3, TypeScript

Cloud & Infrastructure: AWS (Lambda, API Gateway, SQS, SNS, EventBridge, S3), GCP (Cloud Functions, Cloud Storage, BigQuery), Terraform, CDK

Databases: PostgreSQL, MySQL, MongoDB, Couchbase, BigQuery

APIs & Integration: REST, GraphQL, Salesforce APIs, Finicity APIs, SFTP

DevOps & Tools: Docker, Kubernetes, Git (GitHub, GitLab), Nginx, DataDog, SonarQube, Apache Airflow

AI/ML & LLM Stack: OpenAI, Ollama, LangChain, LangGraph, LangFlow, LangFuse, PyTorch, TensorFlow, LLMs, AI Agents, Gemini OCR

Testing: Pytest, Jest, Mocha, Coverage

PROFESSIONAL EXPERIENCE

ACV Auto Auction India Private Limited, Chennai, India

Senior Software Engineer - May 2024 – Present

- Designed and deployed an explainable AI-agent system predicting dealer defaults with **87% accuracy** across 240+ dealers in real time, securing leadership approval for productization. *Python, AI/ML, LLMs, AI Agents*
- Re-architected fragmented Salesforce workflows into a scalable BFF + microservices platform with a centralized risk tracker monitoring **663 active segments at 100% validation accuracy** with real-time DataDog alerting. *Python, AWS (API Gateway, Lambda, S3), Kubernetes, Docker*
- Architected end-to-end NACHA payment automation with **zero manual intervention**, secure SFTP ingestion with validation, retries, and real-time Slack alerting. *Python, AWS, SFTP, NACHA, DataDog, Slack APIs*
- Engineered a mission-critical risk-scoring engine evaluating 20+ financial and behavioral data points for daily credit decisions, with explainable trend visualizations for non-technical stakeholders. *Python, AWS, REST & GraphQL, PostgreSQL, Kubernetes, Docker*
- Integrated Finicity open-banking APIs onboarding 50+ customers and built a Gemini-OCR document-intelligence pipeline that **cut manual review effort by 60%**. *Python, AWS, Finicity APIs, Salesforce, Gemini OCR*

HTC Global Services Private Limited, Chennai, India

Senior Software Engineer - Dec 2018 – May 2024

- Built an end-to-end package-creation and booking platform with a discount engine, seasonal offers, and SES email automation; deployed BFF microservices via Terraform & CDK at **85%+ test coverage**. *Python, AWS, Terraform, PostgreSQL, GraphQL, Angular*
- Delivered an Event Management System spanning virtual, webinar, seminar, and hybrid events across 8 categories with multilingual Contentful CMS; **reduced page load time by 40%** via API optimization and lazy loading. *Python, Node.js, AWS, Terraform, PostgreSQL, GraphQL, Contentful*
- Led a 2-engineer team building an immersive Babylon.js gaming PoC with real-time WebSocket broadcasting, and modernized a legacy Flash-based leadership app with MongoDB/MySQL data design. *Node.js, Express, Socket.IO, MongoDB, MySQL, Babylon.js*
- Built multi-channel chatbots on Yellow Messenger (WhatsApp, Instagram, IVR, Facebook, email) and a full-stack audio-to-text app on AWS Transcribe with profanity filtering. *Python, Node.js, MongoDB, Elastic Search, AWS Transcribe*

Tech Mahindra, Chennai, India

Associate Engineer - May 2017 – Feb 2018

- Documented business processes (Visio, Lucidchart) and performed SQL/Excel data analysis to identify optimization opportunities, collaborating cross-functionally via Jira, Trello, and Confluence.

LEADERSHIP & IMPACT

Team Leadership	Lead a 3-engineer team at ACV - own solutioning, review designs and code, and drive the full software delivery lifecycle from planning through production.
Solution Ownership	Reverse-engineered legacy systems and designed scalable BFF + microservices architectures adopted across credit, risk, and payment workflows.
Business Impact	87% AI prediction accuracy, 100% payment-processing automation, and 60% reduction in manual document review across delivered systems.

PUBLICATIONS

- Sidhaiyan, D., Bhatia, D., & Srinivasan, G. (2026). *GUS: A Graph Utility Score Framework for Method Selection in Fraud Detection*. Proc. 4th Intl. Conf. on Data Science, Agents and Artificial Intelligence (ICDSAAI 2026), Chennai: IEEE.
- Sidhaiyan, D. (2026). *Democratizing Pedagogical Content Creation: A Multi-Agent YouTube AI Framework for Empowering Indian Educators in Higher Education*. Intl. Conf. on Education and Learning with Artificial Intelligence (ICELAI-26).

EDUCATION

B.E., Electrical & Electronics Engineering - Shree Venkateshwara Hi-Tech Engineering College, Tamil Nadu 2013 – 2017